

Languages

- Chinese
- English

Research Interest and Technique

- Soft robotics
- Complaint mechanism
- ✓ Design optimisation
- ✓ Robotic control
- Flexible sensing
- Continuous robot
- ✓ Finite element analysis
- ✓ Kinematics and dynamic analysis

Education

University of Nottingham (United Kingdom)	Research Fellow	2023.11~Now
University of Nottingham (United Kingdom)	PhD in Manufacturing Engineering	2019.09~2024.07
Beihang University (China)	M.Eng. in Mechanical Engineering	2016.09~2019.06
Yantai University (China)	B.Eng. in Mechanical Engineering	2011.09~2015.06

Academic publication

- Published papers (Google Scholar: <https://scholar.google.com/citations?user=qjRL70IAAAAJ&hl=en>)
- [1] **Wang, X.**, Li, S., Chang, J.C., Liu, J., Axinte, D. and Dong, X., 2024. Multimodal locomotion ultra-thin soft robots for exploration of narrow spaces. *Nature Communications*, 15(1), p.6296. (JCR Q1)
 - [2] **Wang, X.**, Liu, Y., Tu, J., Chang, J.C., Wang, F., Axinte, D. and Dong, X., 2025. Bi-stable thin soft robot for in-plane locomotion in narrow space. *IEEE Robotics and Automation Letters*. (JCR Q1)
 - [3] **Wang, X.**, Chang, J.C., Li, S., He, C., Wang, F., Axinte, D. and Dong, X., 2025. Fast-moving thin soft-rigid hybrid robot driven by in-plane dielectric elastomer actuator. *Smart Materials and Structures*. (JCR Q1)
 - [4] **Wang, X.**, Raimondi, L., Axinte, D. and Dong, X., 2025. Investigation on a class of 2D profile amplified stroke dielectric elastomer actuators. *Journal of Mechanisms and Robotics*, 17(3), p.031008. (JCR Q2)
 - [5] **Wang, X.**, Wang, J., Zhang, T., Li, S., Chang, J.C., Zhang, Y., Zeng, T. and Dong, X., 2025. Design and characterisation of the out-of-plane dielectric elastomer generator for wave energy harvesting. *Ocean Engineering*, 339, p.122148. (JCR Q1)
 - [6] **Wang, X.**, Li, S., Chang, J.C., Wang, F. and Dong, X., 2025. Biasing bending DEA-enabled serial kinematic thin soft robot for locomotion and manipulation in confined spaces. 9th International Conference on Robotics, Control and Automation (ICRCA) (pp. 116-121). IEEE.
 - [7] **Wang, X.**, Qian, C., Ma, N. and Dong, X., 2021. Design and Manufacture of an Origami-Inspired Crawling Mini-robot. 14th International Conference on Intelligent Robotics and Applications, *ICIRA* 2021, (pp. 425-436).
 - [8] **Wang, X.**, Chang, J.C., Li, S., and Dong, X., 2025. Design and Fabrication of a Two Degree of Freedom Parallel Robot for Use in Education. 30th International Conference on Automation and Computing (ICAC) (pp. 1-6). IEEE.
 - [9] **Wang, X.**, Chang, J.C., Li, S., Landon, N. and Dong, X., 2025. Design and test of a four revolute joints small-scale robot. 11th International Conference on Optimization and Applications (ICOA) IEEE.
 - [10] Chang, J.C., Dai, H., **Wang, X.**, Axinte, D. and Dong, X., 2025. Development of a continuum robot with inflatable stiffness-adjustable elements for in-situ repair of aeroengines. *Robotics and Computer-Integrated Manufacturing*, 95, p.103018. (JCR Q1)
 - [11] Ba, W., Chang, J.C., Liu, J., **Wang, X.**, Dong, X. and Axinte, D., 2024. An analytical differential kinematics-based method for controlling tendon-driven continuum robots. *Robotics and Autonomous Systems*, 171, p.104562. (JCR Q1)
 - [12] Li, S., **Wang, X.**, Chang, J.C., and Dong, X., 2025. Development and Experimental Evaluation of a Vibration-Based Adhesion System for Miniature Wall-Climbing Robots. 11th International Conference on Optimization and Applications (ICOA) IEEE.

- [13] Dai, H., **Wang, X.**, Crowe, A., Dong, X. and Chen, G., 2024, August. Developing a 3D-Printed Parallel Manipulator for Enhancing Robotics Education. In 2024 29th International Conference on Automation and Computing (*ICAC*) (pp. 1-6).
- [14] Yang, Y. and **Wang, X.**, 2019. Investigation into the linear velocity response of cantilever beam embedded with impact damper. *Journal of Vibration and Control*, 25(7), pp.1365-1378. (*JCR Q2*)
- [15] Wu, D., **Wang, X.** and Yang, Y., 2020. Design and optimisation of a turning cutter embedded with impact damper. *International Journal of Mechatronics and Manufacturing Systems*, 13(3), pp.185-198.
- [16] Yang, Y., Pei, X. and **Wang, X.**, 2018. Dynamics and chatter analysis in robotic milling. *International Journal of Mechatronics and Manufacturing Systems*, 11(1), pp.87-97.

➤ **Papers under review and in preparation**

- [17] Chang, J.C., **Wang, X.**, Axinte, D. and Dong, X. Design, manufacturing, and inverse dynamic modeling of soft parallel robots actuated by dielectric elastomer actuators. arXiv preprint arXiv:2409.20344. (Under review, *IEEE Transactions on Robotics*)
- [18] **Wang, X.**, Dai, H., Li, S., Chang, J.C., Axinte, D. and Dong, X., Reversible pneumatic composite fabric stiffening cell for assisting in-situ inspection of aeroengines (Under review, *Robotics and Computer-Integrated Manufacturing*)
- [19] Li, S., **Wang, X.*** Li, K., Zhang, Z., Kong, X., Chen, G., Zhang, Y., Zeng, T., and Dong, X., Investigation on the lightweight circular ion thruster driven marine vehicle (Under review, *Robotica*)
- [20] Bogatyrev, A.,# **Wang, X.**# Liao, Z., Dong, X., Axinte, D., Laser ablation induced dielectric elastomer driven micro actuator array enable scalable smart surfaces manipulation (Under review, *Soft robotics*)
- [21] **Wang, X.**, Liu, J., Li, S., Dai, H., Chang, J.C., Axinte, D. and Dong, X., Untethered thin dielectric elastomer actuated soft robot. (Under review, *TAROS 2026 | Towards Autonomous Robotic Systems*)
- [22] Chang, J.C., Dai, H., **Wang, X.**, Axinte, D. and Dong, X. Closed-form Evaluation of Insertion-induced Buckling for Continuum Robots: Deployed in a Nuclear Environment. (Under review, *Robotica*)
- [23] **Wang, X.**, Axinte, D. and Dong, X., Soft Robotic Locomotion in Narrow Space Actuation and Functionality (*in preparation*)
- [24] **Wang, X.**, Wild, S., Li, K., Axinte, D. and Dong, X., Integrated perceptual, proximal, and tactile sensing for xxx (*in preparation*)

➤ **Patent**

- [1] **Wang, X.**, Axinte, D. and Dong, X., 2025. "Soft pneumatic composite fabric stiffening system for in-situ robotics" (US Patent, filed) Rolls-Royce Patent Application Ref 2025P00172 GB, application number GB2514683.8.
- [2] **Wang, X.**, Axinte, D. and Dong, X. "Novel tactile sensing for large scale sensor in narrow space" (US Patent, Submitted)

Project and Work Experience

➤ **Project Title:** REPLENISH: Repair and Evaluation of aerospace Products LinE side, Near wing, and In SHop

Project Duration: Jun 2024 – Dec 2027

Funding Source: Innovate UK (UKRI)

Funded Value: £2,184,149

My Responsibilities (Lead Role): Development of a novel pneumatic system for borescopes and continuum robots to perform assisted inspection and repair work on aeroengines. Investigation on the automated soft manipulator for aeroengine inspection by integrating a flexible tactile sensor.

➤ **Project Title:** REINSTATE: Repair, Enhanced Inspection, and Novel Sensing Techniques for increased Availability and reduced Through life Expense

Project Duration: Sep 2020 – Dec 2024

Funding Source: Innovate UK (UKRI)

Funded Value: £1,946,073

My Responsibilities (Lead Role): To address the need for regular inspection in ultra-narrow spaces of new energy-powered aircraft engines, a new type of ultra-thin robot capable of entering these confined spaces has been developed. This robot can perform multimodal movements and execute inspection tasks. The work involves research on smart materials, manufacturing processes, electronic circuits, flexible circuits, mechanical modelling, and simulation analysis. The innovation and future potential of this research were widely recognised at the Rolls-Royce (September 2024) Collaborative R&T for Enhanced Aftermarket Services Conference.

➤ **Project Title:** Developing an 8-meter Disposable Long Snake Manipulator for Surveying Nuclear Environments via Narrow Entry

Project Duration: Apr 2024 – Mar 2025

Funding Source: Robotics and AI Collaboration (RAICo)

Funded Value: £212,000

My Responsibilities (Participate): Proposed the original design of the pneumatic stiffening section for the continuum robot and conducted the first version demonstration.

➤ **Project Title:** EPSRC-IAA - Snake robots for enabling aircraft wing robotic inspection

Project Duration: Jul 2024 – Apr 2025

Funding Source: UK Engineering and Physical Sciences Research Council (EPSRC)

Funded Value: £12,000

My Responsibilities (Lead Role): The project aims to develop continuum robots for inspecting and repairing Airbus A350 aircraft wing fuel tanks. I am responsible for the design, assembly, and testing of the continuum robots.

➤ **Project Title:** EPSRC-IAA - Medical COBRA – Enabling Cadaveric Trials

Project Duration: Sep 2023 – Jun 2024

Funding Source: UK Engineering and Physical Sciences Research Council (EPSRC)

Funded Value: £12,000

My Responsibilities (Lead Role): This project builds upon a previously completed project from our research group (COBRA—Continuum roBot for Remote Applications). In this new phase, a 5-meter-long continuum industrial robot, COBRA, has been adapted for medical applications, now known as "Medical-COBRA." I led the design, manufacturing, and development of the end-effectors (tweezers and hot wire cutter) and the corresponding drive and control systems, as well as project testing.

➤ **Project Title:** CHIMERA - Robotic Inspection of Pressure Vessels

Project Duration: Jan 2019 – Mar 2022

Funding Source: Innovate UK (UKRI)

Funded Value: £370,261

My Responsibilities (Participate): Development of a novel pneumatic system for borescopes and continuum robots to perform assisted inspection and repair work on aeroengines. Investigation on the automated soft manipulator for aeroengine inspection by integrating a flexible tactile sensor.

Honour, Award and Activity

Poster at 2026 IEEE International Conference on Robotics & Automation (Vienna, Austria)	2026.06
Flash talk at 2025 IEEE-RAS International Conference on Soft Robotics (Lausanne, Switzerland)	2025.04
IEEE member	2025-2026
Rolls-Royce Collaborative R&T for Enhanced Aftermarket Services Conference (Rolls-Royce Learning and Development Centre, Derby)	2024
UK-RAS Showcase 12th Sept 2024 (University of Manchester)	2024.09
The Engineer Collaborate to Innovate Awards 2022 (Snakes Everywhere – from aerospace to medical and beyond) (London)	2023.01

Supervision and Teaching Activity

Reviewer of Nature Communications, Journal of Field Robotics, Journal of Mechanical Design, IEEE-RAS International Conference on Soft Robotics (RoboSoft), et.al.	
Soft robot lab student manager in Rolls-Royce University Technology Centre (UTC) in Manufacturing and On-Wing Technology	2022-Now
Co-supervision of MSc project (Cheng He) - Dielectric Elastomer Actuators (DEA) for Soft-robotics	2022.09
Co-supervision of MSc project (Zonglin Li) – In-plane bi-stable mechanism enabled thin dielectric elastomer actuator	2022.09
Co-supervision of two undergraduate interns – Characterisation and optimisation of 3D printed electrostatic adhesive pad	2022.05-2022.10
Co-supervision of MSc project (Zixu Tian) - Dielectric Elastomer Generator (DEG) for wave energy	2024.09
Co-supervision of two undergraduate interns – Flexible tactile sensing for robotic interaction in confined-space inspection	2025.06-2025.09
Co-supervision of BSc project (Yuechen Lin) – Novel tactile sensor for continuum robot	2026.05
Co-supervision of BSc project (Zhiguo Tao) – Flexible sensor for leaf status monitoring	2026.05
Co-supervision of MMME 4085 Group project (Jared Pursglove, James Rose, Poppy Williams, Klejdi Bicprendi, Daanish Salmaan, Aison Howell)	2026.05
Teaching Assistant for Mechanical Design Course at the University of Nottingham	2024-2026

Self-identification

I am an enthusiastic and dedicated academic with a strong passion for robotics, particularly their application in various industries and their impact on human life. Over the course of several years of education and project experiences in the field of engineering, I have developed critical thinking, problem-solving, and collaboration skills that I am eager to apply to complex professional challenges. I am excited to continue my career, engage in impactful research, and collaborate with fellow scholars to push the boundaries of robotics.